

Contents

AI/ML in Biomarker Discovery	1
6-Week Intensive Program — miRNA-Based Biomarker Discovery in Alzheimer's Disease	1
COURSE OVERVIEW	1
Program Philosophy	1
6-Week Syllabus at a Glance	1
Practical Workflow (Runs Across All 6 Weeks)	2
WEEK 1: Foundations in Biomedical Science & Analytical Tools	2
Learning Objectives	2
MODULE 1.1 — Alzheimer's Disease: Biology and the Diagnostic Gap	2
MODULE 1.2 — MicroRNA Biology: From Genome to Biomarker	3
MODULE 1.3 — Literature Review: miRNA Biomarkers in Alzheimer's Disease	6
MODULE 1.4 — Measurement Technologies: How miRNA Data is Generated	8
MODULE 1.5 — Data Repositories & Platforms for miRNA Data	11
MODULE 1.6 — Computational Tools Overview	13
WEEK 1 LAB SESSION: Environment Setup	15
WEEK 1 ASSIGNMENTS	16
WEEK 1 GLOSSARY	16
KEY REFERENCES (Week 1)	17

AI/ML in Biomarker Discovery

6-Week Intensive Program — miRNA-Based Biomarker Discovery in Alzheimer's Disease

COURSE OVERVIEW

Target Audience: Wet-lab biologists with strong biomedical background and little to no coding experience

Duration: 6 weeks, (lectures + hands-on labs)

Disease Focus: Alzheimer's Disease (AD)

Biomarker Class: MicroRNA (miRNA)

Computational Environment: Python (Jupyter notebooks) + R (Bioconductor)

Data Source: Publicly available GEO datasets (blood-based miRNA profiling)

Program Philosophy

This course prioritizes **biological understanding first, computation second**. Every dataset, every algorithm, and every result will be interpreted through the lens of disease biology. You will not become a software engineer; you will become a biologist who can confidently use AI/ML tools and critically evaluate computational analyses in your own research.

6-Week Syllabus at a Glance

Week	Title	Core Topics
1	Foundations: Biology & Analytical Tools	Alzheimer's disease biology, miRNA biology, literature review, measurement technologies, data repositories, tool setup
2	Data Acquisition & Quality Control	Searching GEO, downloading datasets, sample metadata, raw data formats, preprocessing & normalization
3	Exploratory Data Analysis	Descriptive statistics, dimensionality reduction (PCA, t-SNE, UMAP), clustering, visualization, batch effects
4	Feature Selection & Classical ML	Differential expression, feature selection methods, logistic regression, SVM, random forest, model evaluation
5	Advanced ML & Validation	Cross-validation strategies, ensemble methods, deep learning introduction, external validation, overfitting
6	Biological Interpretation & Clinical Translation	Target prediction, pathway enrichment, network analysis, biomarker validation roadmap, regulatory considerations

Practical Workflow (Runs Across All 6 Weeks)

Literature Review → Dataset Selection → Data Download → Quality Control → Normalization → Exploratory Analysis → Differential Expression → Feature Selection → ML Modeling → Biological Interpretation → Clinical Contextualization

WEEK 1: Foundations in Biomedical Science & Analytical Tools

Learning Objectives

By the end of Week 1, you will be able to: 1. Describe the molecular pathology of Alzheimer's disease and explain why current diagnostics are insufficient 2. Explain miRNA biogenesis, function, and why they are attractive biomarker candidates 3. Critically read and discuss primary literature on miRNA biomarkers in AD 4. Distinguish between microarray and next-generation sequencing (NGS) technologies for miRNA profiling 5. Identify major platforms and databases for obtaining miRNA expression data 6. Navigate the key computational tools that will be used throughout the course

MODULE 1.1 — Alzheimer's Disease: Biology and the Diagnostic Gap

1.1.1 What is Alzheimer's Disease?

Alzheimer's disease (AD) is the most common form of dementia, accounting for 60-80% of all dementia cases. It is a progressive, irreversible neurodegenerative disorder characterized by:

- **Amyloid-β (Aβ) plaques** — extracellular aggregates of misfolded amyloid-β peptides derived from the amyloid precursor protein (APP) by sequential cleavage by β-secretase (BACE1) and γ-secretase
- **Neurofibrillary tangles (NFTs)** — intracellular aggregates of hyperphosphorylated tau protein (encoded by the *MAPT* gene), which disrupt microtubule assembly and axonal transport
- **Neuroinflammation** — chronic activation of microglia and astrocytes, driven in part by Aβ accumulation
- **Synaptic loss and neurodegeneration** — ultimately leading to cognitive decline, memory impairment, and loss of function

Key genetic risk factors: - *APOE ε4* allele — most common genetic risk factor for late-onset AD (increases risk 3-12×) - *APP*, *PSEN1*, *PSEN2* mutations — cause rare early-onset familial AD (<1% of cases)

Disease staging: - **Preclinical AD:** Aβ deposits begin accumulating ~15-20 years before symptoms; no cognitive impairment - **Mild Cognitive Impairment (MCI):** Subjective and objective memory complaints; not yet dementia - **Clinical AD:** Progressive dementia affecting daily life

Why this matters for biomarkers: The pathological changes in AD begin **decades before** clinical symptoms appear. Any effective biomarker strategy must detect disease at the preclinical or MCI stage, when disease-modifying interventions would have the greatest impact.

1.1.2 The Diagnostic Challenge

Current gold-standard AD biomarkers include:

Biomarker	Sample Type	Clinical Issue
Aβ42, total tau, p-tau181	Cerebrospinal fluid (CSF)	Invasive lumbar puncture; patient reluctance
Amyloid PET, tau PET	Brain imaging	Expensive (~\$3,000-5,000/scan); limited availability
Clinical cognitive tests (MMSE)	Behavioral	Subjective; only detects established disease
Plasma Aβ42/40, p-tau217	Blood	Emerging; limited clinical infrastructure

The unmet need: A **non-invasive, blood-based biomarker** that can detect AD at preclinical or prodromal stages, is cost-effective, and can be widely deployed in primary care settings. This is precisely the space where **circulating miRNAs** have shown enormous promise.

MODULE 1.2 — MicroRNA Biology: From Genome to Biomarker

1.2.1 What are MicroRNAs?

MicroRNAs (miRNAs) are a class of small (~18-25 nucleotide), single-stranded, non-coding RNA molecules that regulate gene expression at the post-transcriptional level. They were first discovered in 1993 (*lin-4* in *C. elegans*; Lee et al., Science) and recognized as a major regulatory class in 2001.

Key facts: - The human genome encodes ~**2,656 mature miRNAs** (miRBase v22) - Each miRNA can regulate **hundreds to thousands** of target genes - Together, miRNAs are estimated to regulate

>60% of all human protein-coding genes - Named using a standardized nomenclature: e.g., **hsa-miR-21-5p** (hsa = *Homo sapiens*, 21 = gene number, 5p = derived from 5' arm of precursor)

1.2.2 miRNA Biogenesis Pathway

Understanding the biogenesis pathway helps you understand why miRNA expression can go wrong in disease:

NUCLEUS:

Gene (DNA) → RNA Pol II transcription → Primary miRNA (pri-miRNA)

↓
Drosha/DGCR8 complex (Microprocessor)

↓
~70 nt hairpin: Precursor miRNA (pre-miRNA)

↓
Exportin-5 (nuclear export)

CYTOPLASM:

pre-miRNA

↓
Dicer + TRBP complex

↓
~22 bp RNA duplex (miRNA/miRNA* duplex)

↓
Strand selection: one strand (guide) → mature miRNA
other strand (passenger) → degraded

↓
RISC (RNA-Induced Silencing Complex) + AGO2 protein

↓
Target mRNA recognition (partial complementarity, 3' UTR "seed" matching)

↓
mRNA degradation OR translational repression → reduced protein output

Biological consequence: miRNAs act as **rheostats** (fine-tuners), not on/off switches. A single miRNA rarely completely silences a gene; instead, it reduces output by 30–80%, enabling precise modulation of complex biological networks.

1.2.3 Why Are miRNAs ideal biomarker candidates?

miRNAs possess several properties that make them exceptionally attractive as clinical biomarkers:

1. Stability in biofluids Unlike mRNA (which degrades rapidly in blood due to RNases), miRNAs are remarkably stable in serum, plasma, CSF, urine, and saliva. This stability arises from: - Encapsulation within **extracellular vesicles (EVs)** / exosomes - Binding to **Argonaute 2 (AGO2) protein** (protein-protected form) - Association with **HDL/LDL lipoproteins** - Binding to **RNA-binding proteins** (e.g., nucleophosmin)

2. Tissue specificity Many miRNAs are preferentially expressed in specific tissues (e.g., miR-9 and miR-132 are highly enriched in neurons), which means their presence in blood can reflect pathology in the originating tissue.

3. Disease-responsive changes miRNA expression profiles change measurably in response to disease, preceding protein-level or structural changes in many cases.

4. Brain-to-blood communication via EVs Neurons and glial cells release EVs that carry miRNAs into the bloodstream. Some brain-derived EVs can cross or release miRNAs across the

blood-brain barrier (BBB), allowing **non-invasive sampling of brain miRNA profiles through blood draws**.

5. Measurability at scale With current NGS and microarray platforms, hundreds to thousands of miRNAs can be profiled simultaneously from small volumes of biofluids (as little as 200 µL serum).

6. Clinical deployability qPCR-based miRNA assays can be run on standard clinical laboratory equipment, making validated miRNA biomarkers potentially deployable in any hospital or primary care setting.

1.2.4 miRNAs Most Implicated in Alzheimer's Disease Biology

The following miRNAs have been repeatedly identified across AD studies. As a biologist, understanding their *targets* and *pathways* is essential for interpreting computational results:

miRNA	Direction in AD	Key Targets	Biological Relevance
miR-29a/b	↓ Downregulated	BACE1, APP	Decreased miR-29 → increased BACE1 → more Aβ production
miR-107	↓ Downregulated	BACE1, Cofilin-1	One of the earliest changed miRNAs in AD
miR-9	↓ Downregulated (brain)	NFκB, SIRT1	Neuronal miRNA; regulates neuroinflammation
miR-132/212	↓ Downregulated	FOXO3a, tau kinases	Neuroprotective; loss linked to tau hyperphosphorylation
miR-34a	↑ Upregulated	SIRT1, BCL2	Promotes apoptosis, linked to aging
miR-146a	↑ Upregulated	TRAF6, IRAK1	Master regulator of neuroinflammation via TLR/NF-κB
miR-155	↑ Upregulated	SHIP1, C/EBPβ	Neuroinflammatory; elevated in microglia
miR-21-5p	↑ Upregulated	PTEN, PDCD4	Anti-apoptotic; elevated in AD serum
miR-26a/26b-5p	↓ Downregulated	PTEN, CDK5	Strongly correlated with MMSE cognitive scores
miR-532-5p	Variable	Multiple	Highest statistical significance in Ludwig et al. 2019
miR-128	↓ Downregulated	PPARγ, Bax	Neuroprotective; regulates insulin signaling
miR-181	↑ Upregulated	SIRT1, GRP78	Linked to tau phosphorylation and ER stress

Key concept for wet-lab biologists: When a computational analysis identifies a differentially expressed miRNA, your job is to ask: *Which of my target's proteins change? What pathway is perturbed? Is this biologically plausible for AD pathology?* This biological validation thinking is what separates a good bioinformatics study from a great one.

MODULE 1.3 — Literature Review: miRNA Biomarkers in Alzheimer's Disease

The following literature review is based on articles retrieved from PubMed. All citations include DOI links.

1.3.1 Early Biomarker Discovery and the Case for Blood-Based Approaches

The quest for minimally invasive AD biomarkers has driven significant innovation. An influential early review by Zafari et al. (2015) *Gerontology* (DOI: [10.1159/000375236](https://doi.org/10.1159/000375236)) outlined the landscape of circulating biomarker panels in AD, contrasting established CSF markers (A β 42, tau) with emerging blood-based approaches. The authors highlighted the critical need for high-throughput profiling technologies — microarrays and next-generation sequencing — to discover miRNA signatures capable of detecting AD non-invasively.

Zetterberg and Burnham (2019) *Molecular Brain* (DOI: [10.1186/s13041-019-0448-1](https://doi.org/10.1186/s13041-019-0448-1)) provided a comprehensive review of blood-based molecular biomarkers for AD, situating miRNAs within a broader landscape that includes plasma A β , tau, and neurofilament light chain (NfL). They emphasized that **blood-based biomarkers are critical for primary care screening** — the clinical setting where most patients first present — and that a scalable, cost-effective blood test for AD represents one of the field's most pressing unmet needs.

1.3.2 CSF miRNA Profiling

Cerebrospinal fluid, being in direct contact with the brain parenchyma, was among the first biofluids interrogated for miRNA biomarkers. Müller et al. (2015) *PLoS One* (DOI: [10.1371/journal.pone.0126423](https://doi.org/10.1371/journal.pone.0126423)) applied **OpenArray technology** — a medium-throughput qPCR-based approach capable of profiling 1,178 unique miRNAs simultaneously — to CSF from AD patients and controls. This landmark study identified several miRNAs specifically dysregulated in AD CSF, including members of the let-7 family and miR-29 family, and demonstrated that **CSF miRNA signatures can distinguish AD from other dementias**. However, the invasiveness of lumbar puncture limits the clinical applicability of CSF-based approaches.

1.3.3 Blood-Based miRNA Biomarker Discovery

The majority of recent biomarker discovery work has shifted to blood (serum/plasma), which can be obtained via routine venipuncture.

A landmark systematic review by Fattahi et al. (2024) *Metabolic Brain Disease* (DOI: [10.1007/s11011-024-01431-7](https://doi.org/10.1007/s11011-024-01431-7)) synthesized findings from 48 studies comprising **4,001 AD patients and 3,886 healthy controls**. Their key findings: - **83 miRNAs were consistently upregulated** in AD blood - **66 miRNAs were consistently downregulated** in AD blood - Whole blood (39.6%) and serum (27.1%) were the most common sample types - No single miRNA emerged as universally consistent — **panel-based approaches** (multiple miRNAs together) consistently outperform single miRNA biomarkers - RT-qPCR was the most commonly used validation method

Wang et al. (2023) *Int J Mol Sci* (DOI: [10.3390/ijms242216259](https://doi.org/10.3390/ijms242216259)) provided a mechanistic review connecting circulating miRNA profiles to core AD pathology. They catalogued miRNAs that directly regulate: - **MAPT** (tau gene): miR-9, miR-132, miR-212, miR-34a regulate tau phosphorylation and splicing - **APP** (amyloid precursor protein): miR-101, miR-16, miR-153 suppress APP translation - **BACE1** (β -secretase, the enzyme that cleaves APP to produce A β): miR-29 family, miR-107, miR-339-5p — **arguably the most therapeutically relevant miRNA-target axis in AD**

This mechanistic grounding is essential: differentially expressed miRNAs in blood are not merely correlation signals; they often reflect **real causal biology** happening in the brain.

1.3.4 Extracellular Vesicles: A Bridge Between Brain and Blood

A crucial recent advance concerns **neuronal-derived extracellular vesicles (NDEVs)**. Reho et al. (2025) *Alzheimer's & Dementia* (DOI: [10.1002/alz.70050](https://doi.org/10.1002/alz.70050)) conducted a transcriptome-wide association study of NDEV-derived miRNAs in serum from 46 clinical AD patients, 14 preclinical AD patients, and 60 healthy controls — a multiethnic cohort.

Key findings: - **14 miRNAs** were significantly associated with AD risk - Preclinical AD individuals showed **more pronounced transcriptional alterations** than clinical AD individuals — suggesting NDEV miRNAs may be particularly powerful for early detection - Key target genes identified: **SNCA** (alpha-synuclein), **CYCS** (cytochrome c), and **MAPT** (tau) — all central to neurodegeneration

This study illustrates the **power of combining EV biology with miRNA profiling**: by isolating neuron-specific EVs, researchers can enrich for brain-derived signals in a peripheral blood sample, effectively giving us a “liquid biopsy” of the brain.

Liquid biopsy concept: A review by Malhotra et al. (2023) *Cells* (DOI: [10.3390/cells12141911](https://doi.org/10.3390/cells12141911)) defines liquid biopsy as the analysis of disease-relevant biological molecules — including miRNAs, cell-free DNA, and EVs — from non-solid biofluids (blood, urine, CSF, saliva). In neurological diseases, liquid biopsy is emerging as a transformative approach precisely because traditional tissue biopsy of the brain is not feasible in living patients.

1.3.5 Machine Learning Applied to miRNA Biomarker Discovery

The application of ML to miRNA biomarker panels has substantially improved diagnostic accuracy over individual miRNAs. This is the central technical focus of our course.

Ludwig et al. (2019) *Genomics, Proteomics & Bioinformatics* (DOI: [10.1016/j.gpb.2019.09.004](https://doi.org/10.1016/j.gpb.2019.09.004)) is a landmark study. Starting from high-throughput sequencing-derived miRNA signatures in US and German cohorts, the authors validated 21 circulating miRNAs in **465 individuals** (AD patients + controls) using RT-qPCR. A machine learning model achieved an **AUC of 87.6%** for AD vs control classification — a clinically meaningful performance threshold. Critically, **miR-26a/26b-5p** showed significant correlation with MMSE scores (cognitive severity), and **miR-532-5p** showed the highest statistical significance in disease association. miRNAs downregulated in AD were enriched in monocytes and T-helper cells; those upregulated were enriched in serum exosomes, suggesting cell-type-specific release mechanisms.

Zhao et al. (2020) *J Applied Lab Medicine* (DOI: [10.1373/jalm.2019.029595](https://doi.org/10.1373/jalm.2019.029595)) profiled >500 miRNAs in 96 serum samples using multiplex RT-qPCR (OPTIMA cohort). A **random forest** classifier identified a **12-miRNA signature** achieving 76–85.7% accuracy for AD diagnosis. Notably, the signature accuracy improved substantially when validated on post-mortem histology-confirmed AD cases (85.7%), highlighting the challenge of clinical diagnosis variability.

Xu et al. (2022) *J Alzheimer's Disease* (DOI: [10.3233/JAD-215502](https://doi.org/10.3233/JAD-215502)) built ML models using not just miRNA expression values but also **target gene descriptors and pathway features** derived from known miRNA-target interactions. Their best serum-based model achieved **92.0% accuracy** and their plasma-based model achieved **90.9% accuracy**. This study demonstrates a critical concept: incorporating **biological knowledge** (target gene networks, pathways) into ML feature engineering can substantially improve model performance — a strategy we will implement in Week 4 of this course.

Viswambharan et al. (2017) *Progress in Molecular Biology and Translational Science* (DOI: [10.1016/bs.pmbts.2016.12.013](https://doi.org/10.1016/bs.pmbts.2016.12.013)) reviewed miRNAs as peripheral biomarkers in aging and age-related diseases, contextualizing AD miRNA findings within the broader landscape of aging biology and discussing the stabilizing mechanisms (exosomal packaging, protein binding) that make circulating miRNAs measurable.

1.3.6 Summary: What the Literature Tells Us

Finding	Implication for Course
No single miRNA biomarker is sufficient	We will build panel-based ML classifiers , not single-marker tests
Panel performance peaks at 10–30 miRNAs	Feature selection (Week 4) is critical to avoid overfitting
Performance varies by sample type (serum vs plasma vs whole blood)	Metadata management and batch correction (Week 2) are essential
Biological knowledge improves ML performance	We will use target gene and pathway features alongside expression values
Preclinical detection is the frontier	Our ML models will include MCI samples as a distinct class
External validation is rarely done	We will implement cross-cohort validation using multiple GEO datasets

MODULE 1.4 — Measurement Technologies: How miRNA Data is Generated

Understanding how your data was generated is fundamental to interpreting it correctly. Two major technologies dominate miRNA profiling: **microarray** and **next-generation sequencing (small RNA-seq)**.

1.4.1 Technology 1: Microarray-Based miRNA Profiling

Principle:

Microarrays use **hybridization** — the property of complementary nucleic acid strands to bind to each other. Probes (short oligonucleotides complementary to known miRNA sequences) are spotted or synthesized in fixed positions on a glass or silicon chip. Labeled sample RNA is hybridized to the chip; signal intensity at each probe position reflects the abundance of the corresponding miRNA.

Experimental workflow:

```

Biofluid collection (serum/plasma)
    ↓
RNA extraction (Qiagen miRNeasy, Thermo TRIzol-LS)
    ↓
RNA quality check (Bioanalyzer or TapeStation)
    ↓
Sample labeling (Cy3/Cy5 fluorescent dyes, or biotin)
    ↓
Hybridization to microarray chip (12–18 hours, 55°C)
    ↓
Chip washing & scanning (laser scanner)
    ↓
Image analysis → signal intensity matrix
    ↓
Normalization → Expression matrix (samples × miRNAs)

```

Major microarray platforms for miRNA:

Platform	Company	# miRNAs	Notes
GeneChip miRNA 4.0 Array	Affymetrix/Thermo Fisher	~2,578 human miRNAs	Gold standard for sensitivity; also detects pre-miRNAs
SurePrint G3 Human miRNA 8x60K	Agilent Technologies	~2,006 human miRNAs	High dynamic range; dual-color design
Human miRNA Expression BeadChip	Illumina	~1,532 human miRNAs	Bead-based; high reproducibility
TaqMan Array Human MicroRNA A+B cards	Applied Biosystems	754 miRNAs/card	qPCR-based array; highest sensitivity
OpenArray Human miRNA Panel	Applied Biosystems	754 or 1,178 assays	Used in Müller et al. 2015 CSF study

Strengths of microarray: - Well-established; large body of published data in GEO for cross-study comparison - Cost-effective for large sample sizes ($n > 50$) - Quantification is rapid (~24 hours) - Simpler bioinformatic pipeline - No need for library preparation

Limitations of microarray: - **Closed vocabulary:** Can only detect miRNAs with existing probes; novel miRNAs are missed - **Cross-hybridization:** Probes with imperfect complementarity can bind related miRNAs (false signals) - **Lower dynamic range** compared to sequencing - **Probe design bias:** Detection efficiency varies by miRNA sequence composition - **GC-content effects** on hybridization efficiency require careful normalization

Data output: A matrix of fluorescence intensity values (or log2-transformed ratios) for each miRNA $\times$ sample combination. Typical output: hundreds to thousands of miRNA probes $\times$ N samples.

1.4.2 Technology 2: Small RNA Sequencing (miRNA-seq)

Principle:

Small RNA sequencing uses **next-generation sequencing (NGS)** to directly sequence small RNA molecules from a sample. Unlike microarrays, it is not limited to known sequences — any small RNA present in sufficient abundance will be detected and quantified.

Experimental workflow:

```

Biofluid collection
↓
RNA extraction (specialized small RNA extraction: Qiagen miRNeasy,
                NEB Monarch, or exosome isolation first)
↓
Size selection (gel electrophoresis or bead-based: select 18–30 nt fragments)
↓
3' adapter ligation (RNA ligase)
↓
5' adapter ligation
↓
Reverse transcription (cDNA synthesis)
↓

```

PCR amplification (add sample index/barcode for multiplexing)
↓
Library quality check (Bioanalyzer: ~145 bp expected product)
↓
Pooling & sequencing (Illumina short-read platform, 50 bp single-end)
↓
Raw FASTQ files (millions of short reads per sample)
↓
Adapter trimming → Alignment to genome/miRBase →
Read counting → Count matrix (samples × miRNAs)

Major sequencing platforms used for miRNA-seq:

Platform	Company	Read Length	Throughput	Common Use
NovaSeq 6000/X	Illumina	50-150 bp	Very high (6 Tb)	Population-scale studies
HiSeq 2500/4000	Illumina	50-150 bp	High	Standard research; many GEO datasets
MiSeq	Illumina	150-300 bp	Low	Smaller studies, clinical validation
NextSeq 500/550	Illumina	75-150 bp	Medium	Mid-scale studies
Ion Torrent S5	Thermo Fisher	Variable	Medium	Alternative to Illumina

Why 50 bp single-end reads? miRNAs are ~22 nt; with 3' adapter sequence (~30 nt), total insert size is ~52 bp. Single-end 50 bp reads are sufficient and cost-effective. Paired-end sequencing offers no advantage for miRNA-seq.

Key bioinformatics steps for miRNA-seq (conceptual overview for Week 1):

1. **Adapter trimming:** Remove the 3' adapter sequence added during library prep (using tools like Trim Galore or Cutadapt)
2. **Alignment:** Map trimmed reads to the human genome (hg38) or directly to miRBase hairpin sequences (using tools like miRDeep2, STAR, or Bowtie)
3. **Quantification:** Count reads mapping to each annotated miRNA (using featureCounts or miRDeep2)
4. **Output:** A **count matrix** — integer counts of reads per miRNA per sample

Strengths of small RNA-seq: - **Open discovery:** Can detect novel miRNAs and isomiRs (miRNA sequence variants) - **Higher dynamic range** — can detect very low and very high abundance miRNAs simultaneously - **Single-base resolution** — can distinguish highly similar miRNA family members - **Simultaneous profiling** of other small RNAs (piRNAs, snoRNAs, tRFs) from the same data - Unbiased — no probe design needed

Limitations of small RNA-seq: - **More expensive** per sample (library prep + sequencing costs) - **More complex bioinformatics** — requires alignment, reference genomes, multiple QC steps - **Ligation bias** — adapter ligation efficiency varies by miRNA sequence, affecting quantification accuracy - **Longer turnaround time** - **Requires more input RNA** quality consciousness (low integrity RNA causes bias)

1.4.3 Technology 3: RT-qPCR (Validation Gold Standard)

While not a primary discovery platform, RT-qPCR remains the **gold standard for validation** of candidate biomarkers identified by microarray or sequencing.

Principle: Reverse transcription of miRNA into cDNA, followed by quantitative PCR amplification using miRNA-specific primers. Signal is quantified as cycle threshold (Ct) — lower Ct = higher expression.

Major platforms: - **TaqMan miRNA Assays (Applied Biosystems):** Stem-loop RT primer captures specific miRNA; TaqMan probe provides high specificity. Industry standard. - **miRCURY LNA miRNA PCR Assays (Qiagen):** LNA-modified primers improve hybridization; no separate RT step needed for each miRNA. - **SYBR Green-based assays:** Lower cost but less specific.

When used in a biomarker pipeline: - **Discovery phase:** Microarray or miRNA-seq (high-throughput, discovery) - **Screening phase:** OpenArray or TaqMan Array Cards (medium-throughput, 96–754 miRNAs) - **Validation phase:** RT-qPCR on individual candidates (low-throughput, highest confidence)

1.4.4 Comparison Summary

Feature	Microarray	Small RNA-seq	RT-qPCR
Throughput	High (1,000s miRNAs)	High (all known + novel)	Low (1–96/run)
Dynamic range	Limited (~3 logs)	Wide (~5 logs)	Wide (~5–6 logs)
Novel miRNA detection	No	Yes	No
Sensitivity	Moderate	High	Highest
Specificity	Moderate (cross-hybridization risk)	High (sequence-based)	Very high
Cost per sample	Moderate	High	Low (per target)
Bioinformatics complexity	Low	High	Very low
Data in GEO for AD	Abundant	Growing	Limited (summary stats)
Typical use	Discovery	Discovery / Deep profiling	Validation

Course data: We will primarily work with **GEO-deposited microarray and RNA-seq datasets** from AD studies. Understanding these technologies helps you critically evaluate study design, understand normalization choices, and correctly interpret QC metrics.

MODULE 1.5 — Data Repositories & Platforms for miRNA Data

1.5.1 Primary Data Repositories

1. NCBI GEO (Gene Expression Omnibus)

Website: <https://www.ncbi.nlm.nih.gov/geo/>

- World's largest public repository for functional genomics data - Contains both raw and processed data for microarray and NGS experiments - Searchable by disease, tissue, organism, platform, and publication - Data is organized into: - **GSE** (Series) — a complete study submission - **GSM** (Samples) — individual sample records - **GPL** (Platforms) — the array or sequencing platform used - **GDS** (Datasets) — curated, analysis-ready subsets

Key AD miRNA datasets in GEO for our course:

GEO Accession	Study Focus	Sample Type	Platform	N Samples
GSE46579	AD blood miRNA	Whole blood	Affymetrix	~65
GSE67491	AD serum miRNA	Serum	Affymetrix	~150
GSE63501	AD plasma miRNA	Plasma	Affymetrix	~80
GSE120584	AD serum miRNA (sequencing)	Serum	Illumina	~200
GSE57353	AD brain miRNA	Brain tissue	Illumina	~100

We will work with one or more of these datasets across Weeks 2-5.

2. EBI ArrayExpress

Website: <https://www.ebi.ac.uk/arrayexpress/>

- European counterpart to GEO; many studies cross-deposited in both - Strict MIAME (Minimum Information About a Microarray Experiment) compliance - Good source for validation datasets when looking for independent cohorts

3. miRBase — The miRNA Registry

Website: <https://www.mirbase.org/>

- **Official repository for miRNA sequences, annotations, and nomenclature** - Current release: v22.1 (2588 hairpin precursors, 2656 mature sequences for *H. sapiens*) - Every miRNA referenced in the literature has a miRBase accession number (e.g., MI0000077 for hsa-miR-21) - Essential for: looking up sequences, understanding biogenesis (5p vs 3p arms), finding synonyms from older nomenclature

4. ADNI (Alzheimer's Disease Neuroimaging Initiative)

Website: <http://adni.loni.usc.edu/>

- Multi-site longitudinal study with deep clinical, imaging, and biofluid data - Includes blood-based omics data with corresponding clinical phenotypes (MMSE, CDR, amyloid PET) - Requires data access application (free for academic researchers) - Invaluable for linking miRNA changes to cognitive severity and disease progression

5. AMP-AD Knowledge Portal (via Synapse)

Website: <https://adknowledgeportal.synapse.org/>

- Data from the Accelerating Medicines Partnership — Alzheimer's Disease program - Includes ROSMAP (Religious Orders Study and Memory and Aging Project): deep multi-omic profiling of brain tissue from ~1,000 individuals, including miRNA data - Also includes Mount Sinai Brain Bank and Mayo Clinic RNAseq Study data

1.5.2 Supporting Databases for Biological Interpretation

These databases will be used in Weeks 5-6 for interpreting computational findings:

Database	URL	Purpose
miRTarBase	https://mirtarbase.cuhk.edu.cn	Experimentally validated miRNA-target interactions
TargetScan	https://www.targetscan.org	Computational miRNA target predictions
miRDB	https://mirdb.org	Machine learning-based target predictions
miRNet	https://www.mirnet.ca	miRNA-target network visualization and enrichment
DIANA-miRPath	https://diana.e-ce.uth.gr/mir-pathv3	KEGG/GO pathway analysis for miRNA sets

Database	URL	Purpose
miRWalk	https://mir-walk.umm.uni-heidelberg.de	Comprehensive target database (3' UTR, CDS, 5' UTR)
KEGG	https://www.kegg.jp	Pathway maps for interpreting target gene lists
STRING	https://string-db.org	Protein-protein interaction networks

MODULE 1.6 — Computational Tools Overview

This section introduces the tools we will use throughout the course. No coding is expected this week — the goal is to become familiar with the landscape and understand what each tool is for.

1.6.1 Programming Languages

Python

- General-purpose language dominant in ML/AI applications - We use it for: data manipulation, machine learning models, visualization - Key libraries: - pandas — data tables (like Excel, but for code) - numpy — numerical operations - scikit-learn — machine learning algorithms - matplotlib / seaborn — plotting and visualization - scipy — statistical tests - biopython — biological sequence tools

R

- Statistical computing language dominant in genomics/bioinformatics - We use it for: differential expression analysis, Bioconductor packages - Key packages (from Bioconductor): - DESeq2 — differential expression for count data (RNA-seq) - limma — differential expression for microarray or voom-transformed RNA-seq - edgeR — differential expression using negative binomial models - ggplot2 — publication-quality visualization

Why both? The field uses both languages. Most ML pipelines are Python-first; most genomics preprocessing pipelines are R-first. We will use each where it excels and provide templates so you are never writing code from scratch.

1.6.2 Development Environment

Jupyter Notebook / JupyterLab

- Interactive computational notebook where code, results, and explanatory text coexist - Every code cell can be run individually — excellent for step-by-step learning - We will provide **pre-built notebooks** for each week's lab session

RStudio

- Integrated development environment (IDE) for R - Contains a script editor, console, variable inspector, and plot viewer - We will use R Markdown documents — similar to Jupyter but for R

1.6.3 R Packages for miRNA-Specific Analysis

Package	Purpose	Install Command
GE0query	Download data directly from NCBI GEO	<code>BiocManager::install("GE0query")</code>
DESeq2	Differential expression for RNA-seq count data	<code>BiocManager::install("DESeq2")</code>
limma	Differential expression for microarray data	<code>BiocManager::install("limma")</code>
multiMiR	Integrates 14 miRNA-target databases	<code>BiocManager::install("multiMiR")</code>
miRBaseConverter	Convert between miRNA name versions	<code>BiocManager::install("miRBaseConverter")</code>
clusterProfiler	Gene ontology and KEGG enrichment	<code>BiocManager::install("clusterProfiler")</code>
pheatmap	Heatmap visualization	<code>install.packages("pheatmap")</code>
ggplot2	Publication-quality plots	<code>install.packages("ggplot2")</code>

1.6.4 Python Packages for ML Analysis

Package	Purpose	Install Command
pandas	Data manipulation and analysis	<code>pip install pandas</code>
numpy	Numerical operations	<code>pip install numpy</code>
scikit-learn	ML algorithms (SVM, RF, logistic regression)	<code>pip install scikit-learn</code>
matplotlib	Base plotting library	<code>pip install matplotlib</code>
seaborn	Statistical visualization	<code>pip install seaborn</code>
scipy	Statistical tests (Mann-Whitney, t-test)	<code>pip install scipy</code>
umap-learn	UMAP dimensionality reduction	<code>pip install umap-learn</code>
shap	SHAP values for ML model interpretation	<code>pip install shap</code>
statsmodels	Statistical modeling	<code>pip install statsmodels</code>

1.6.5 The Full Computational Workflow (Preview)

Week 2: R

GE0query → Download raw data (CEL files or count matrix)

limma / DESeq2 → Normalization and quality control

Week 3: Python + R

pandas/numpy → Data structuring and exploration

ggplot2/seaborn → Visualization (boxplots, PCA, heatmaps)

umap-learn → Dimensionality reduction

Week 4: Python

scikit-learn → Differential expression + feature selection

scikit-learn → Logistic Regression, SVM, Random Forest
ROC curves, AUC, cross-validation → Model evaluation

Week 5: Python

sklearn.ensemble → Advanced ensemble methods
SHAP → Feature importance interpretation
External dataset → Cross-cohort validation

Week 6: R + Python

multiMiR → miRNA-target lookup
clusterProfiler → Pathway enrichment analysis
miRNet / Cytoscape → Network visualization

WEEK 1 LAB SESSION: Environment Setup

Objective: By end of lab, every student has a working Python and R environment capable of running the course notebooks.

Lab 1A — Setting Up Python (Anaconda)

1. Download and install **Anaconda** from <https://www.anaconda.com/download>
2. Create a dedicated course environment:

```
conda create -n biomarker_ml python=3.11
conda activate biomarker_ml
pip install pandas numpy scikit-learn matplotlib seaborn scipy umap-learn shap statsmodels jupyter
```

3. Launch JupyterLab:

```
jupyter lab
```

4. Run the verification notebook provided (verifies all packages installed correctly)

Lab 1B — Setting Up R and RStudio

1. Download and install **R** from <https://cran.r-project.org/>
2. Download and install **RStudio** from <https://posit.co/download/rstudio-desktop/>
3. Open RStudio and run:

```
install.packages("BiocManager")
BiocManager::install(c("GEOquery", "DESeq2", "limma", "edgeR",
                      "multiMiR", "clusterProfiler", "pheatmap"))
install.packages(c("ggplot2", "tidyverse", "readr", "dplyr"))
```

4. Verify installation:

```
library(GEOquery)
library(DESeq2)
library(ggplot2)
cat("All packages loaded successfully!\n")
```

WEEK 1 ASSIGNMENTS

Reading Assignment (Required)

1. **Fattahi et al. (2024)** — *Blood-based microRNAs as potential biomarkers for Alzheimer's disease: systematic review* (DOI: [10.1007/s11011-024-01431-7](https://doi.org/10.1007/s11011-024-01431-7))
Focus on: Table of dysregulated miRNAs, sample type comparison, Figure summarizing consistency across studies
2. **Ludwig et al. (2019)** — *Machine Learning to Detect Alzheimer's Disease from Circulating Non-coding RNAs* (DOI: [10.1016/j.gpb.2019.09.004](https://doi.org/10.1016/j.gpb.2019.09.004))
Focus on: Study design, which miRNAs were validated, how the ML model was built and evaluated

Reflection Questions (Discuss in Week 2 opening session)

1. Why do you think no single miRNA has emerged as a reliable standalone AD diagnostic? What biological and technical factors contribute to this?
2. The Ludwig et al. study used RT-qPCR for validation rather than sequencing. What advantages does this offer for translating the findings to a clinical test?
3. If you were designing a miRNA biomarker study for AD today, would you use serum or plasma? What are the key considerations?

Optional Deep Dive

- Reho et al. (2025) (DOI: [10.1002/alz.70050](https://doi.org/10.1002/alz.70050)) — *Preclinical Alzheimer's disease shows alterations in circulating neuronal-derived extracellular vesicle microRNAs*
This is cutting-edge 2025 work combining EV biology with miRNA profiling — an excellent preview of where the field is heading.

WEEK 1 GLOSSARY

Term	Definition
miRNA	MicroRNA; small (~22 nt) non-coding RNA that regulates gene expression post-transcriptionally
pri-miRNA	Primary miRNA transcript; first product of miRNA gene transcription
pre-miRNA	Precursor miRNA; ~70 nt hairpin produced by Drosha/DGCR8 cleavage
RISC	RNA-Induced Silencing Complex; protein complex (with AGO2) that uses mature miRNA to find and silence target mRNAs
Isomer (isomiR)	Sequence variant of a canonical miRNA; same general sequence but with 1-3 nt differences at 5' or 3' end
Extracellular vesicle (EV)	Membrane-enclosed particle (30-5,000 nm) released by cells into biofluids; carries proteins, nucleic acids, lipids
Exosome	Small EV (30-150 nm) of endosomal origin; carries high miRNA cargo
Liquid biopsy	Sampling and analysis of disease biomarkers (miRNA, ctDNA, CTCs) from body fluids
MCI	Mild Cognitive Impairment; prodromal stage of AD with objective memory impairment but preserved daily function
BACE1	Beta-site APP Cleaving Enzyme 1; the β -secretase that initiates A β production from APP

Term	Definition
MMSE	Mini-Mental State Examination; standard clinical cognitive scoring (0–30; <24 indicates impairment)
GEO	Gene Expression Omnibus; NCBI database for genomic expression data
AUC	Area Under the (ROC) Curve; measure of classifier performance (0.5 = random, 1.0 = perfect)
Ct value	Cycle threshold in qPCR; the PCR cycle at which signal crosses threshold; lower Ct = higher expression
Normalization	Mathematical procedure to remove systematic technical variation between samples before biological comparison
CEL file	Raw data file format for Affymetrix microarrays; contains probe-level fluorescence intensities
FASTQ	Raw data format for NGS; text file containing read sequences and per-base quality scores

KEY REFERENCES (Week 1)

All references retrieved from PubMed.

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13. Jaber SA et al. (2021). Novel Plasma miRNAs as Biomarkers and Therapeutic Targets of Alzheimer's Disease at the Prodromal Stage. *J Alzheimers Dis* 83(2):1063-1080. DOI: [10.3233/JAD-210307](https://doi.org/10.3233/JAD-210307)

Next Week: Data Acquisition & Quality Control — We will download a real AD miRNA dataset from GEO, examine its structure, perform quality control, and normalize the data for downstream analysis.